MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, INVERTER, DIGITAL, KEY CODE EMM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inverter gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0929 failures per million operating hours for class I or 0.5138 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EMM.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/261-1 for class I and M28787/261-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

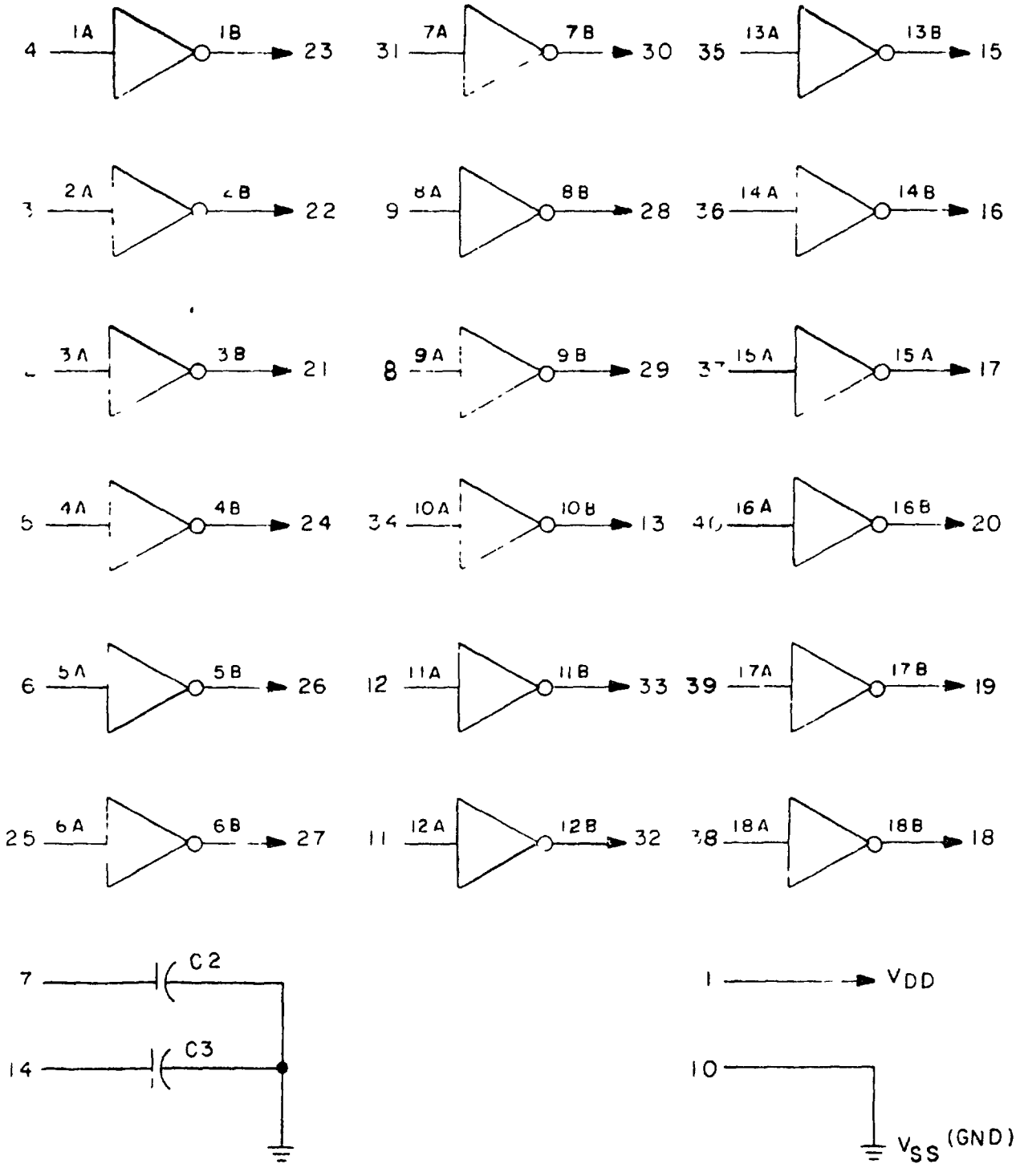
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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C2 = C3 = 220 pF ±20%.

FIGURE 1. EIM module function diagram, inverter gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	Input clamp diode voltage, negative	III	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C18	Input clamp diode voltage, positive	IV	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D18	Output voltage, high level, I_{OH} = 0	V	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E18	Output voltage, low level, I_{OL} = 0	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F18	Output voltage, high level, V_{DD} = 5 V	VII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G18	Output voltage, low level, V_{DD} = 5 V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H18	Output voltage, high level, V_{DD} = 15 V	IX	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
J1-J18	Output voltage, low level, V_{DD} = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K18	Input current, high level	XI	X	0	200	---	---	0	200	---	---	0	200	nA
L1-L18	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
M1	Power supply current, low level	XIII	X	0	6	0	6	0	60	0	6	0	120	µA
M2	Power supply current, high level	XIII	X	0	6	0	6	0	60	0	6	0	120	µA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XIV	S	20	200	20	200	22	220	15	150	24	245	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XIV	Q	68	680	68	680	75	750	51	515	83	835	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XV	S	20	200	20	200	22	220	15	150	24	245	ns
P19-P36	Transition time (t_{THL}),	XV	Q	68	680	68	680	75	750	51	515	83	835	ns

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Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XVI	Q	10	100	10	100	11	110	7	75	12	125	ns
R1-R18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XVII	Q	10	100	10	100	11	110	7	75	12	125	ns
S1-S18	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XVIII	Q	8	80	8	80	9	90	6	60	10	100	ns
T1-T18	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XIX	Q	8	80	8	80	9	90	6	60	10	100	ns
U1-U18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XX	S	12	125	12	125	14	140	9	95	15	155	ns
U19-U36	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	32	325	32	325	36	360	24	245	40	400	ns
V1-V18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	12	125	12	125	14	140	9	95	15	155	ns
V19-V36	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	32	325	32	325	36	360	24	245	40	400	ns
W1-W18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	8	80	8	80	9	90	6	60	10	100	ns
X1-X18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIII	Q	8	80	8	80	9	90	6	60	10	100	ns
Y1-Y18	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	6	60	6	60	6	65	4	45	7	75	ns
Z1-Z18	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXV	Q	6	60	6	60	6	65	4	45	7	75	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA	Power supply filter capacitance	XXVI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1	C2 capacitance	XXVII	X	176	224	176	224	176	224	176	224	176	224	pF
BB2	C3 capacitance	XXVII	X	176	224	176	224	176	224	176	224	176	224	pF
CC1-CC18	Input capacitance	XXVIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows. A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J, M through Z, and CC shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions K and L shall be performed at 25°, 60°, and 100°C only using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. For the transient thermal tests, use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1.8 max	1.8 max	mW
Input current (V_{DD} = 18 V)			
Low level (V_{IL} = 0 V)	0 Min, -200 Max	0 Min, -200 Max	nA
Low level (V_{IL} = 18 V)	0 Min, 200 Max	0 Min, 200 Max	nA
Output drive current (V_{DD} = 5 V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 Min	mA
at 0° and 25°C	0.51 Min	0.51 Min	mA
at 60°C	0.44 Min	0.44 Min	mA
at 100°C	---	0.36 Min	mA
High level (V_{OH} = 4.6 V)			
I_{OH} at -55°C	---	-0.64 Min	mA
at 0° and 25°C	-0.51 Min	-0.51 Min	mA
at 60°C	-0.44 Min	-0.44 Min	mA
at 100°C	---	-0.36 Min	mA

6.5 Module function. The module contains eighteen independent inverter gates and two 220 pF capacitors with one side grounded for external use.

6.5.1 Parallel outputs. Paralleling of outputs is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 1, 2, 3, 4, 5, 6
b. 7, 8, 9, 10, 11, 12
c. 13, 14, 15, 16, 17, 18

6.6 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula.

$$P_{Total} = (CV_{DD}{}^2 f)N + \text{quiescent power dissipation}$$

where: C = Load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

6.7 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedure should be followed:

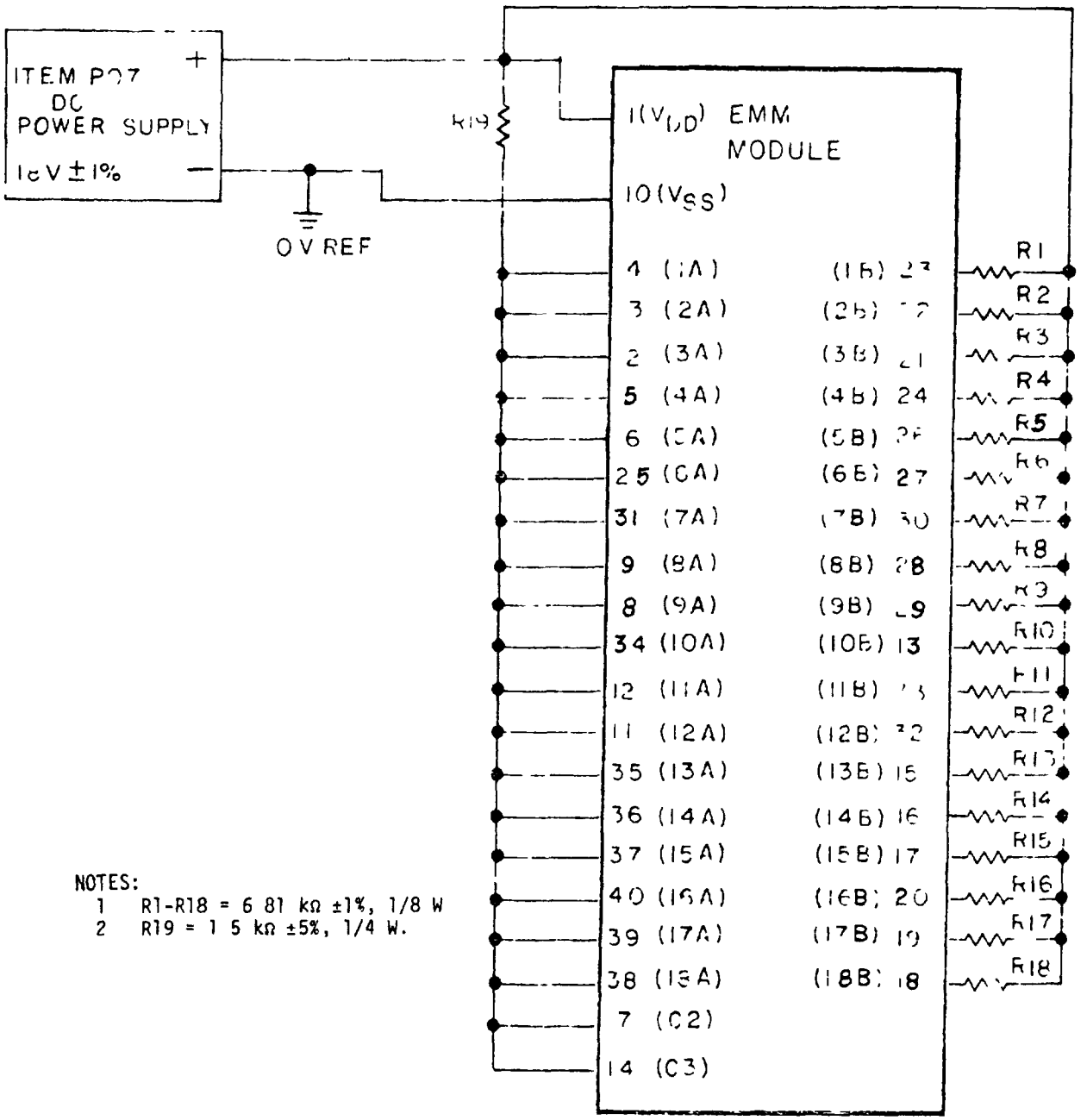
a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.
b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply is off.
d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.
e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the input protection diodes.
f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

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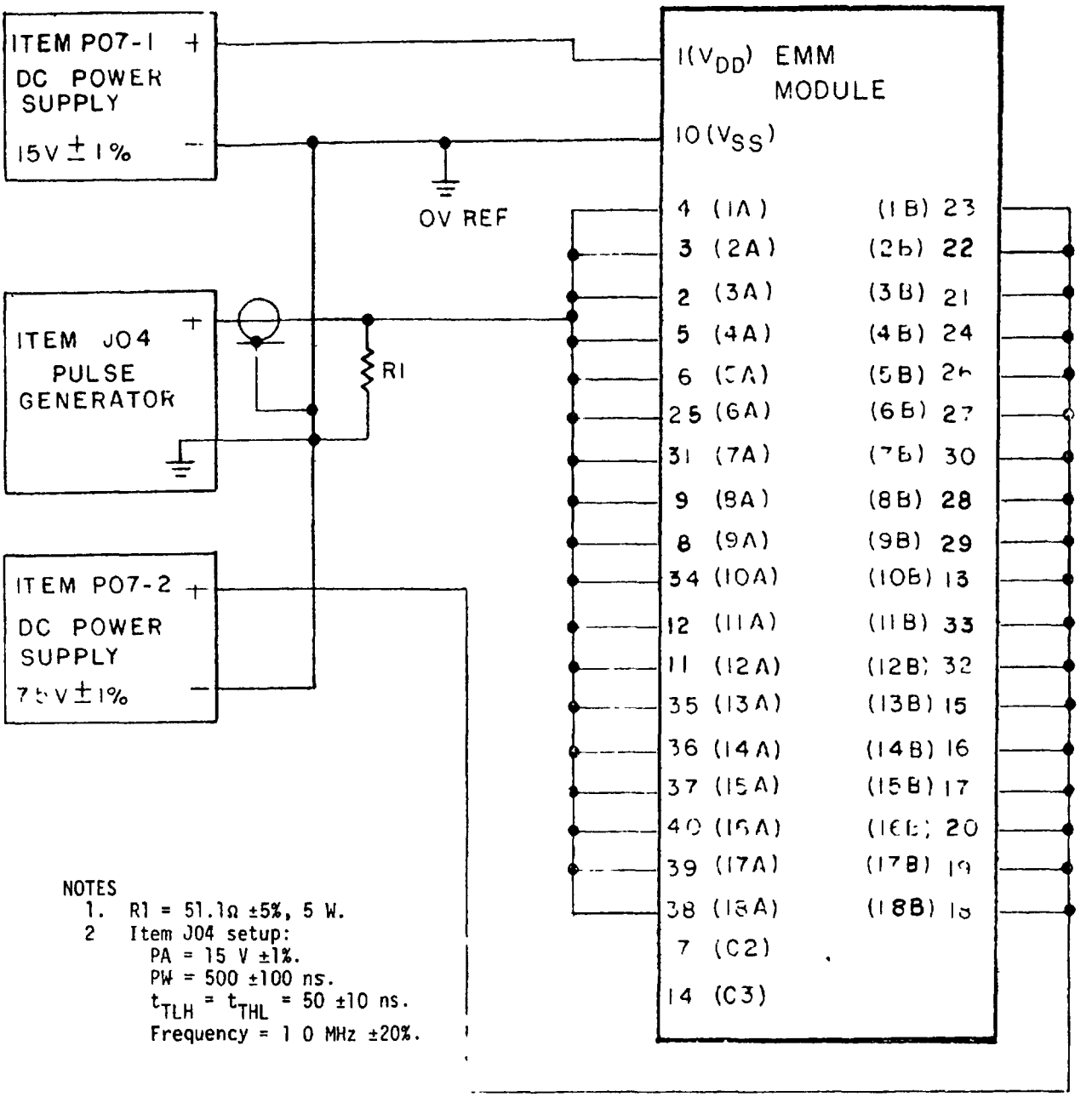
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NOTES:
1. R1-R18 = 6.81 kΩ ±1%, 1/8 W
2. R19 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES
1. R1 = 51.1Ω ±5%, 5 W.
2 Item J04 setup:
PA = 15 V ±1%.
PW = 500 ±100 ns.
$t_{TLH} = t_{THL}$ = 50 ±10 ns.
Frequency = 1 0 MHz ±20%.

FIGURE 3 Component temperature test circuit

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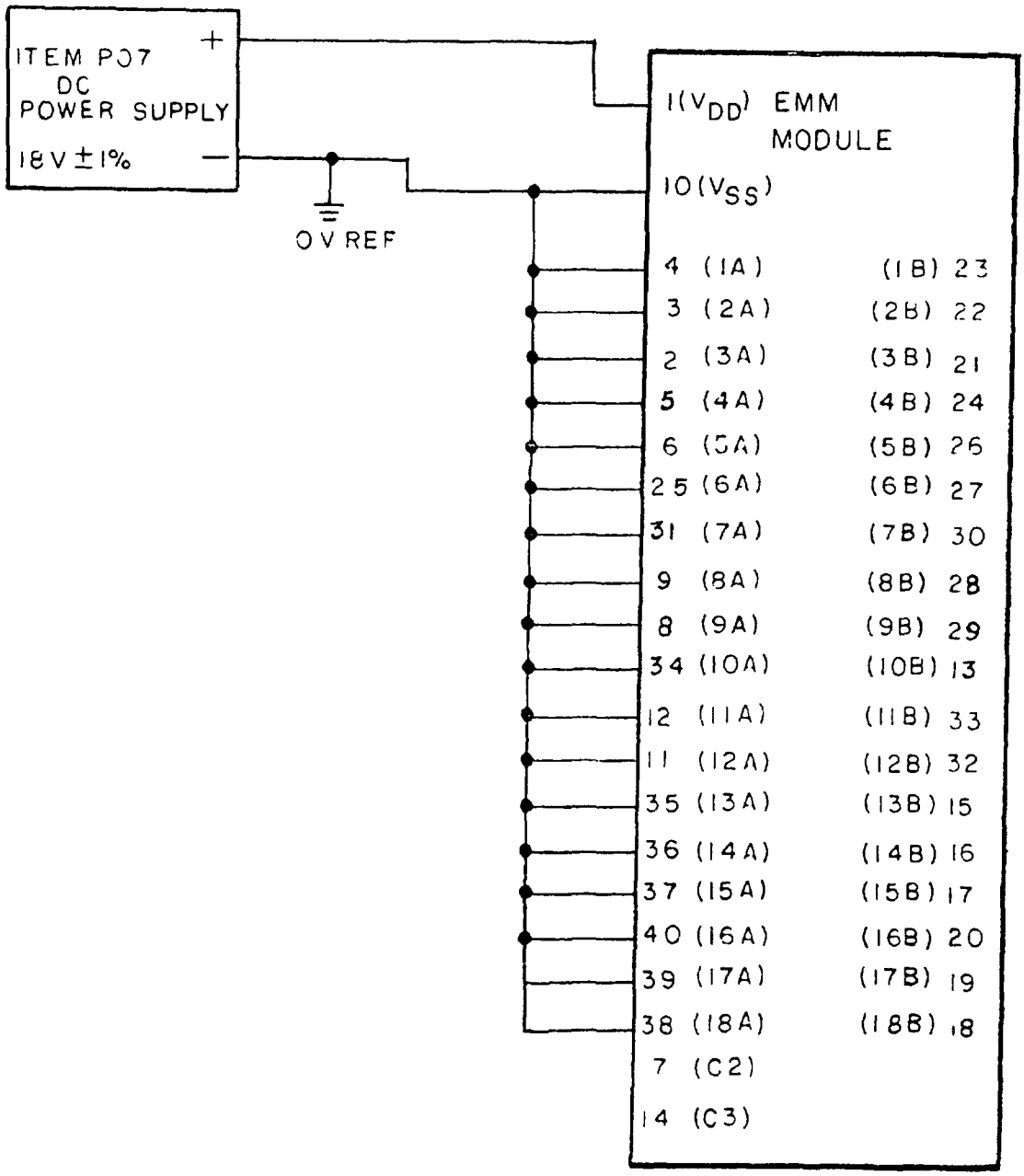


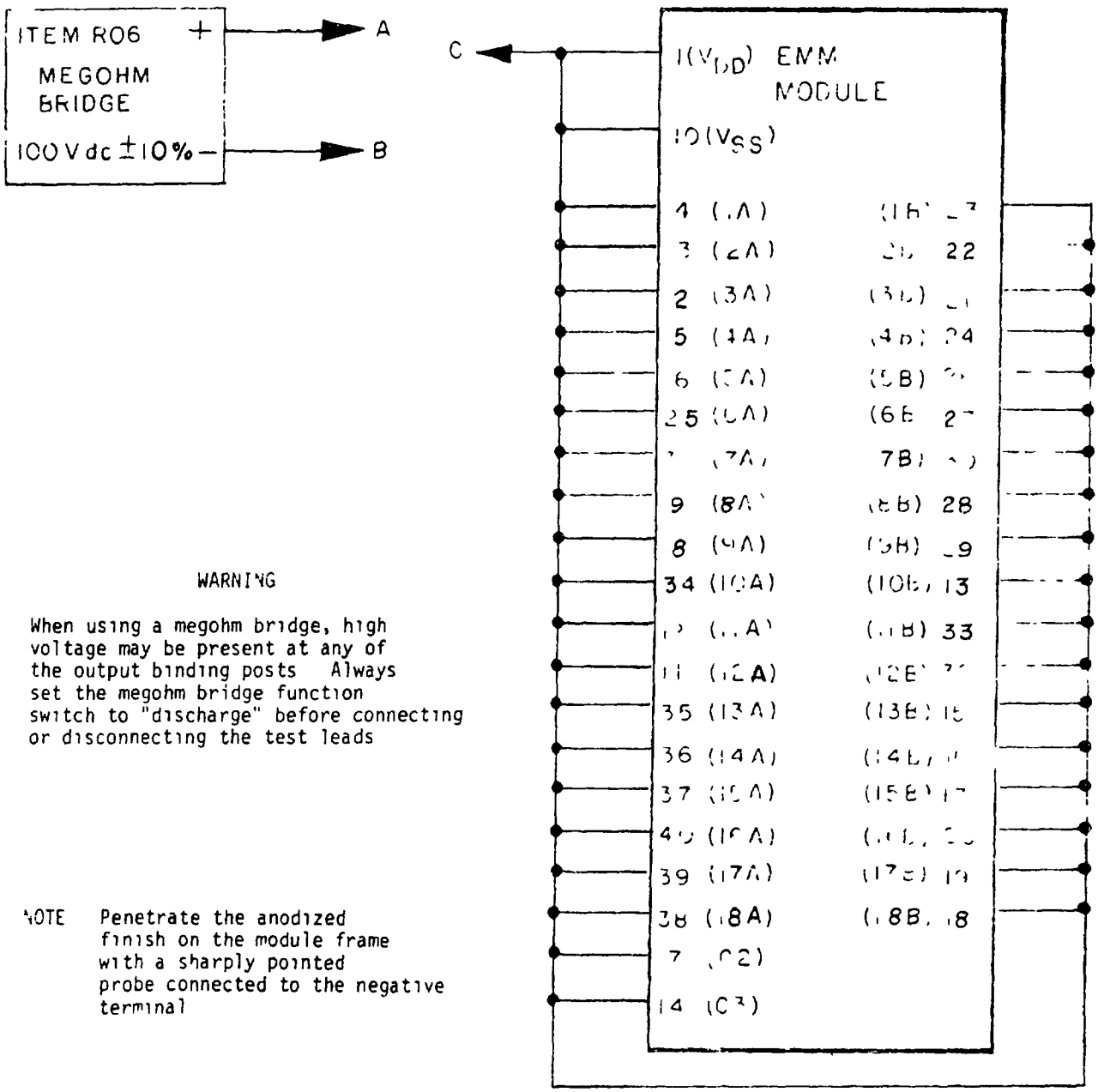
FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	1Ω	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 5. Insulation test circuit.

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TABLE III. Input clamp diode voltage (negative) test.

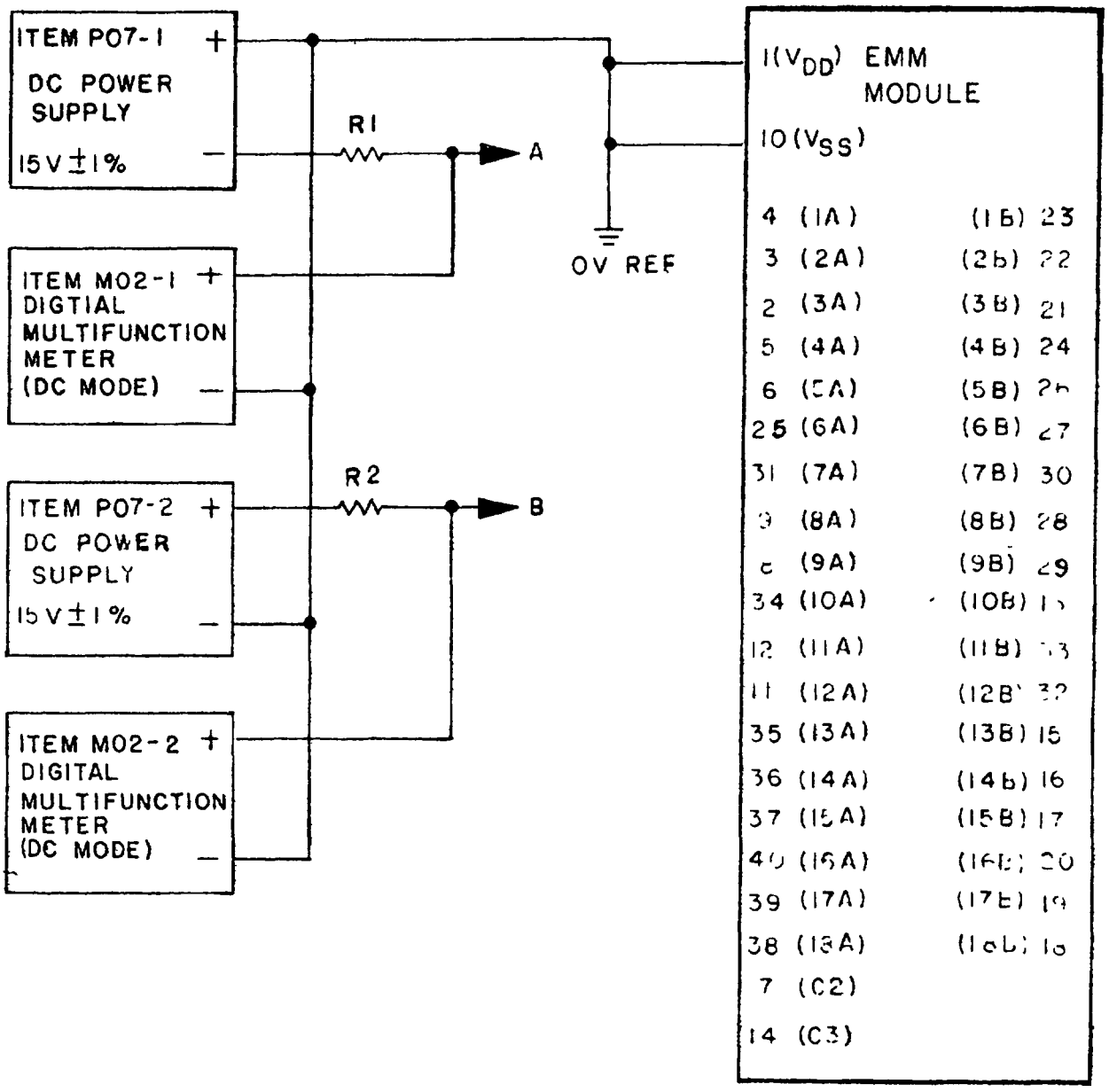
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage, negative	4	Open	-0.4	-6.0	V	6
B2	2A	3					
B3	3A	2					
B4	4A	5					
B5	5A	6					
B6	6A	25					
B7	7A	31					
B8	8A	9					
B9	9A	8					
B10	10A	34					
B11	11A	12					
B12	12A	11					
B13	13A	35					
B14	14A	36					
B15	15A	37					
B16	16A	40					
B17	17A	39					
B18	18A	38					

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TABLE IV. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage, positive	Open	4	0.4	1.5	V	6
C2	2A		3				
C3	3A		2				
C4	4A		5				
C5	5A		6				
C6	6A		25				
C7	7A		31				
C8	8A		9				
C9	9A		8				
C10	10A		34				
C11	11A		12				
C12	12A		11				
C13	13A		35				
C14	14A		36				
C15	15A		37				
C16	16A		40				
C17	17A		39				
C18	18A		38				

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1, P07-2, M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for table III and IV.

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TABLE V. Output voltage (high-level, I_{OH} = 0) test.

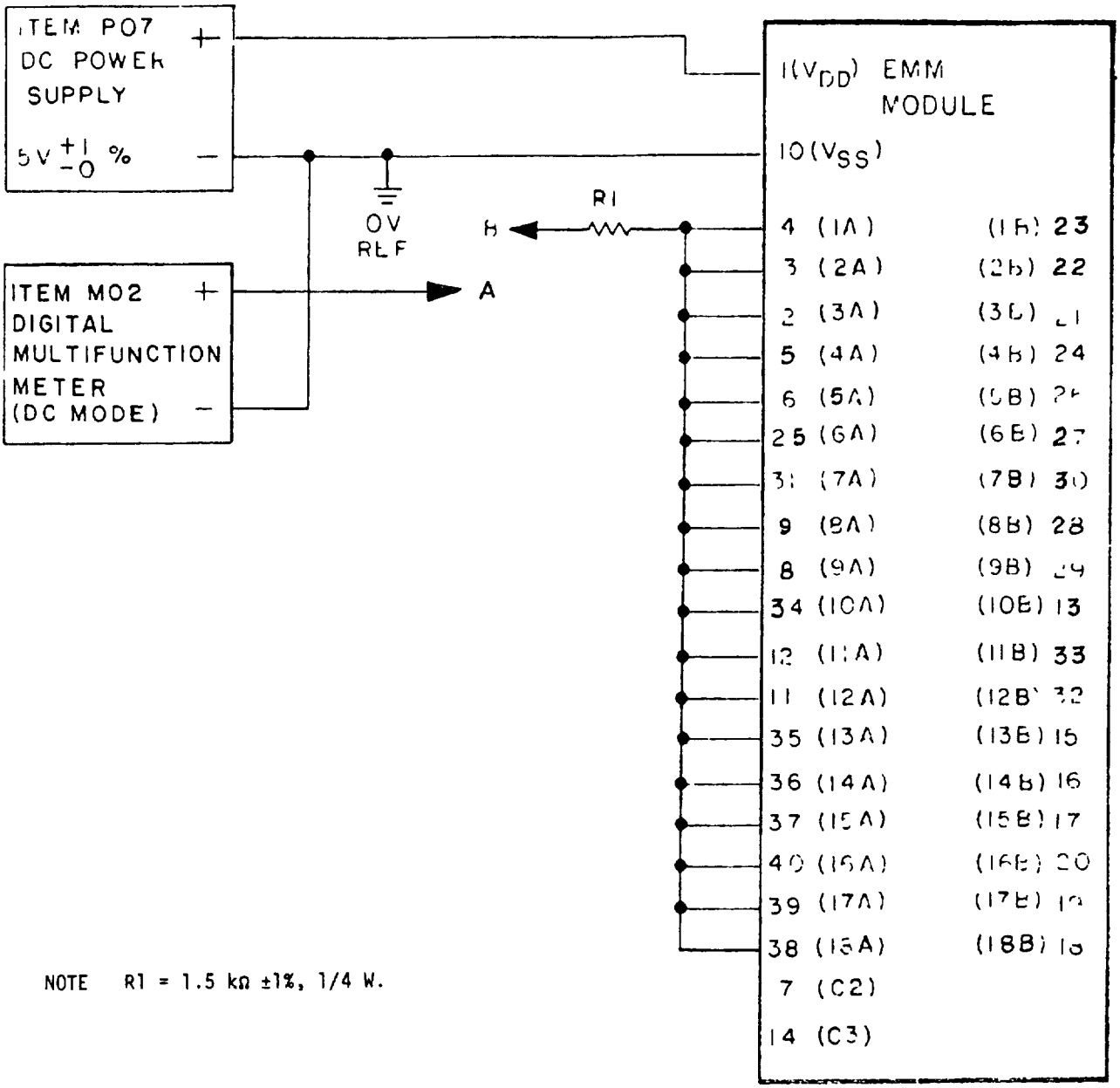
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1B Output voltage, high level, I_{OH} = 0 V	23	10	4.95	5.05	V	7
D2	2B	22					
D3	3B	21					
D4	4B	24					
D5	5B	26					
D6	6B	27					
D7	7B	30					
D8	8B	28					
D9	9B	29					
D10	10B	13					
D11	11B	33					
D12	12B	32					
D13	13B	15					
D14	14B	16					
D15	15B	17					
D16	16B	20					
D17	17B	19					
D18	18B	18					

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TABLE VI. Output voltage (low-level, I_{OL} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1B Output voltage, low level, I_{OL} = 0 V	23	1	0.00	0.05	V	7
E2	2B	22					
E3	3B	21					
E4	4B	24					
E5	5B	26					
E6	6B	27					
E7	7B	30					
E8	8B	28					
E9	9B	29					
E10	10B	13					
E11	11B	33					
E12	12B	32					
E13	13B	15					
E14	14B	16					
E15	15B	17					
E16	16B	20					
E17	17B	19					
E18	18B	18					

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NOTE R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 7. Output voltage (high-level and low-level) test circuit for tables V and VI.

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TABLE VII. Output voltage (high-level, V_{DD} = 5 V) test.

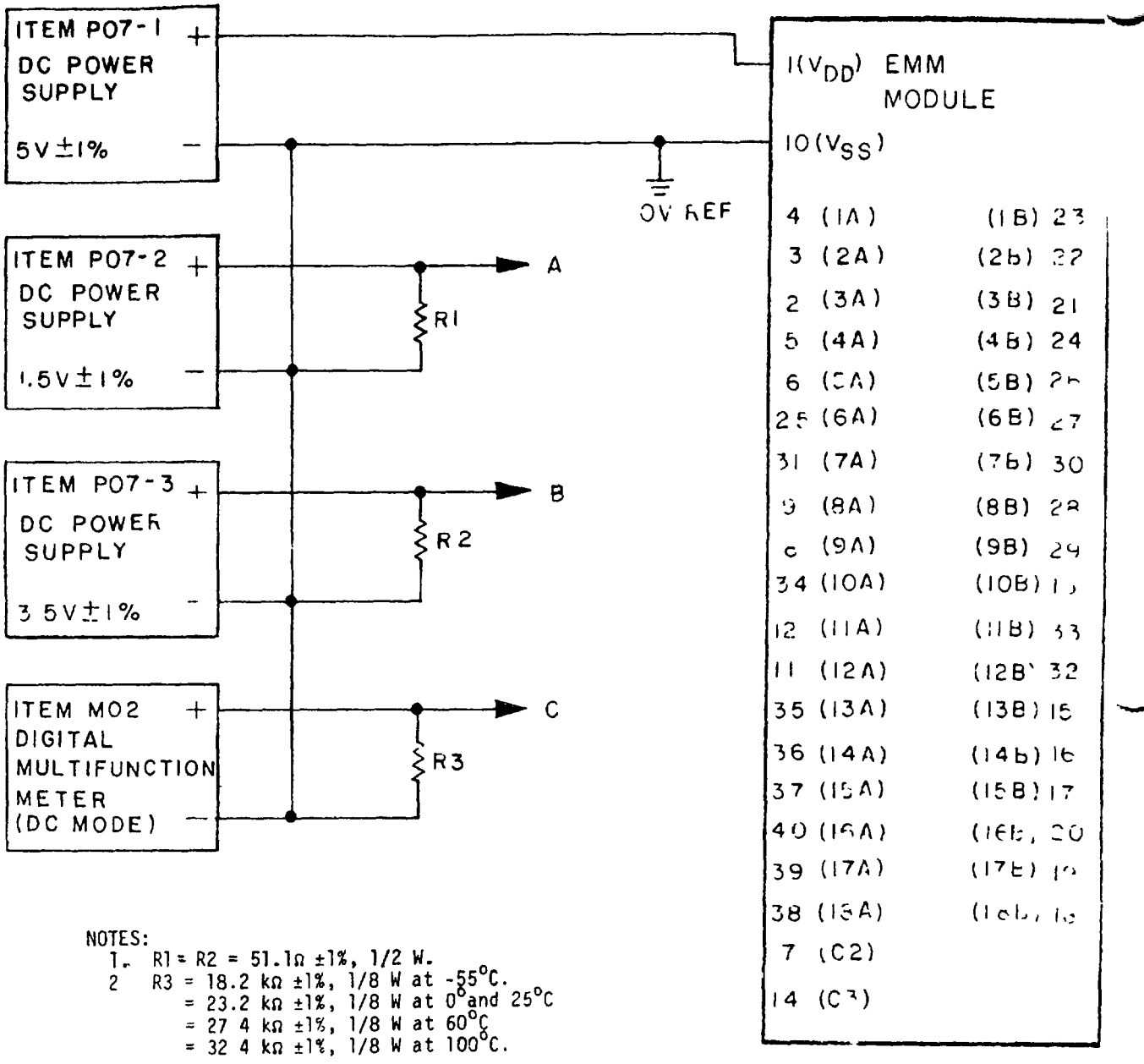
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1B Output voltage, high level, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	4.6	5.0	V	8
F2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
F3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
F4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
F5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
F6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
F7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
F8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
F9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
F10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
F11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
F12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				
F13	13B	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15				
F14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				

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TABLE VII. Output voltage (high-level, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F15	15B Output voltage, high level, V_{DD} = 5 V	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17	4.6	5.0	V	8
F16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20	↓	↓	↓	↓
F17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
F18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/8 W at -55°C.
= 23.2 kΩ ±1%, 1/8 W at 0° and 25°C
= 27.4 kΩ ±1%, 1/8 W at 60°C
= 32.4 kΩ ±1%, 1/8 W at 100°C.

FIGURE 8. Output voltage (high-level, V_{DD} = 5 V) test circuit for table VII.

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TABLE VIII. Output voltage (low-level, V_{DD} = 5 V) test.

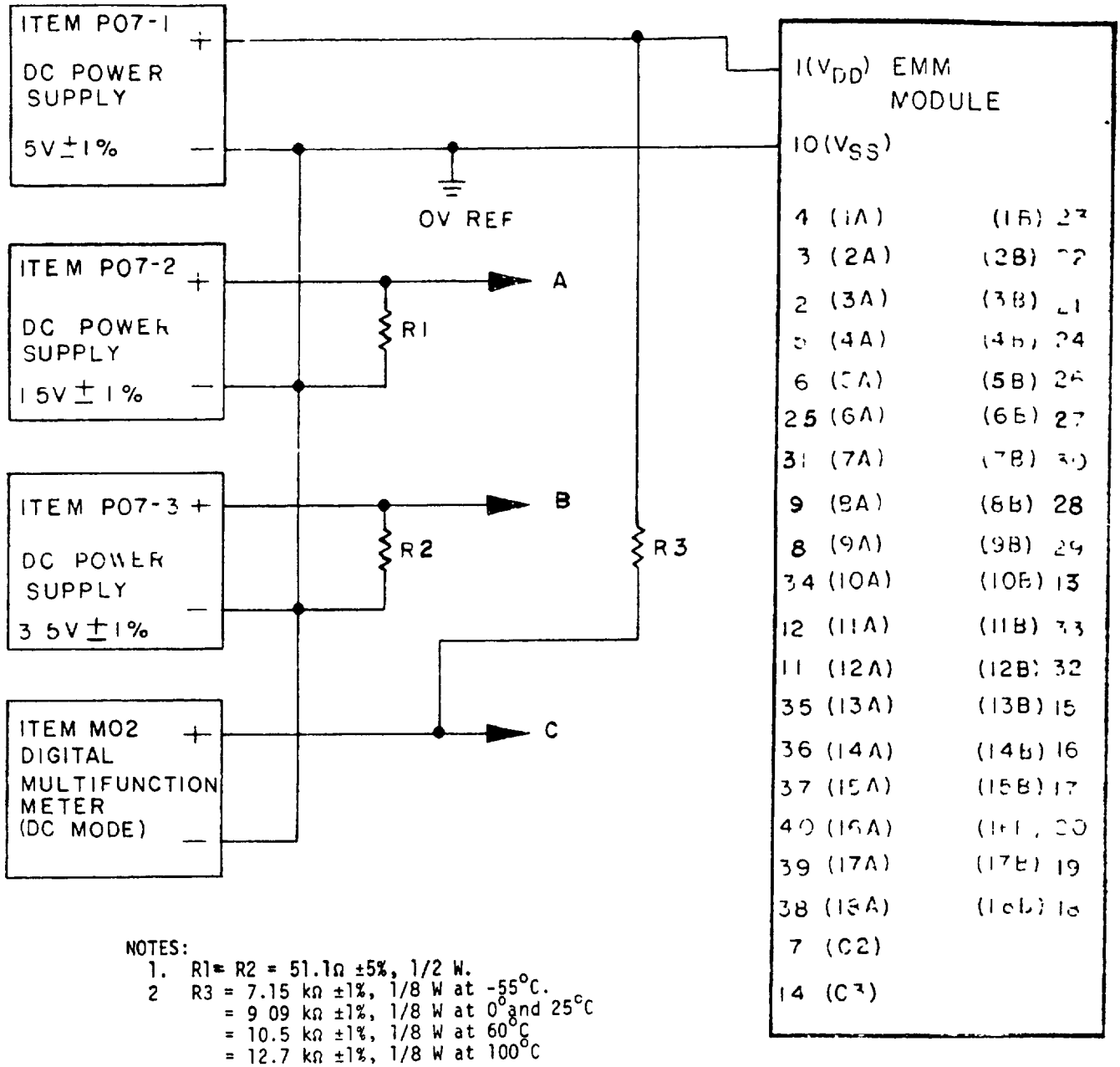
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1B Output voltage, low level, V_{DD} = 5 V	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	4	23	0.0	0.4	V	9
G2	2B	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	3	22				
G3	3B	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	2	21				
G4	4B	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	5	24				
G5	5B	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	6	26				
G6	6B	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	25	27				
G7	7B	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	31	30				
G8	8B	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	9	28				
G9	9B	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	8	29				
G10	10B	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	34	13				
G11	11B	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	12	33				
G12	12B	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	11	32				
G13	13B	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	35	15				
G14	14B	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	36	16				

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TABLE VIII. Output voltage (low-level, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G15	15B Output voltage, low level, V_{DD} = 5 V	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	37	17	0.0	0.4	V	9
G16	16B	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	40	20				
G17	17B	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	39	19				
G18	18B	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	38	18				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 7.15 kΩ ±1%, 1/8 W at -55°C.
= 9.09 kΩ ±1%, 1/8 W at 0° and 25°C
= 10.5 kΩ ±1%, 1/8 W at 60°C
= 12.7 kΩ ±1%, 1/8 W at 100°C

FIGURE 9. Output voltage (low-level, V_{DD} = 5 V) test circuit for table VIII.

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TABLE IX. Output voltage (high-level, V_{DD} = 15 V) test.

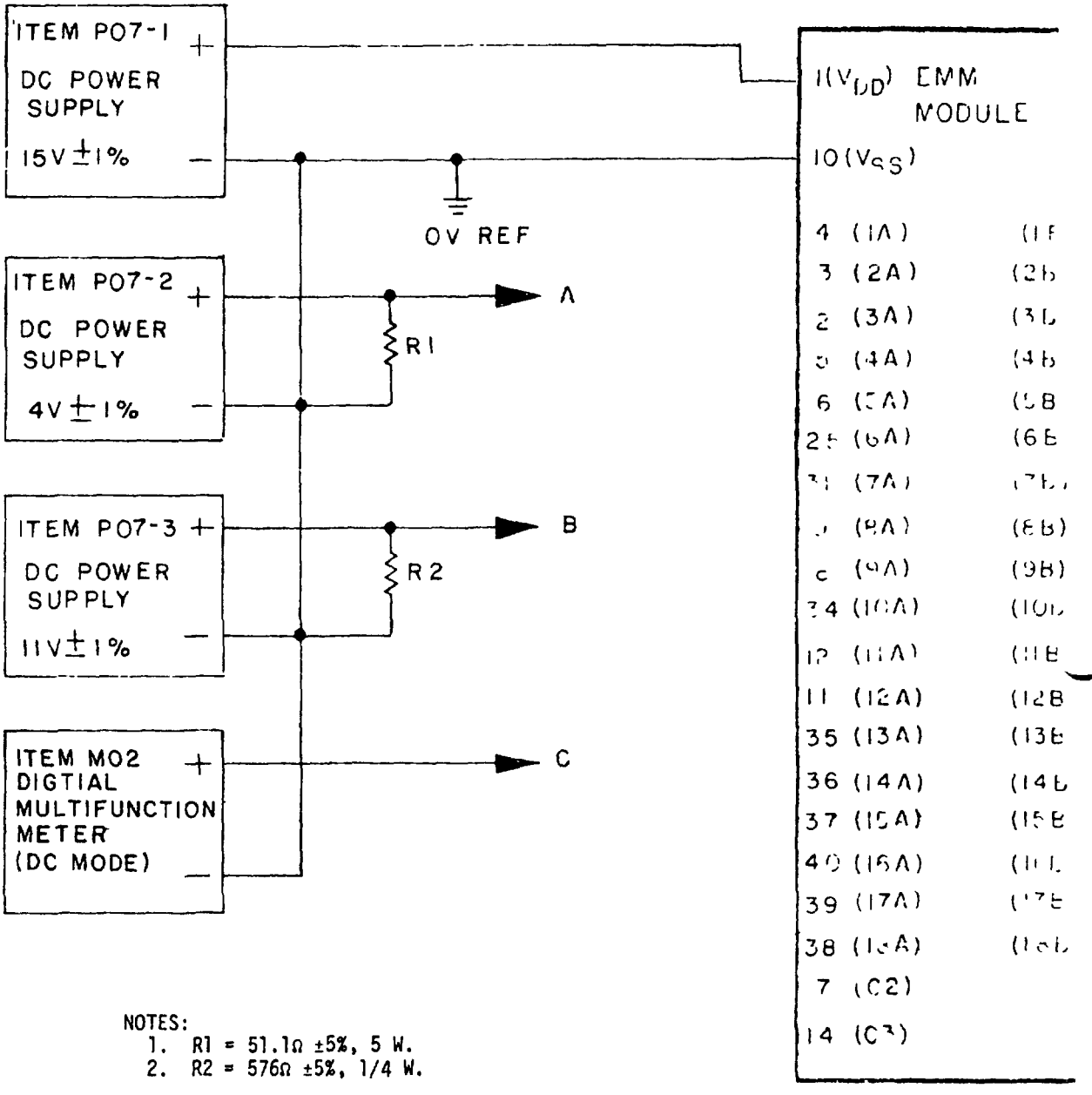
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Output voltage, high level, V_{DD} = 15 V	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39,38,	Open	23	13.5	15.0	V	10
H2	2B			22				
H3	3B			21				
H4	4B			24				
H5	5B			26				
H6	6B			27				
H7	7B			30				
H8	8B			28				
H9	9B			29				
H10	10B			13				
H11	11B			33				
H12	12B			32				
H13	13B			15				
H14	14B			16				
H15	15B			17				
H16	16B			20				
H17	17B			19				
H18	18B			18				

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TABLE X. Output voltage (low-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1B Output voltage, low level, V_{DD} = 15 V	Open	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39,38	23	0.0	1.5	V	10
J2	2B			22				
J3	3B			21				
J4	4B			24				
J5	5B			26				
J6	6B			27				
J7	7B			30				
J8	8B			28				
J9	9B			29				
J10	10B			13				
J11	11B			33				
J12	12B			32				
J13	13B			15				
J14	14B			16				
J15	15B			17				
J16	16B			20				
J17	17B			19				
J18	18B			18				

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NOTES:
1. R1 = 51.1Ω ±5%, 5 W.
2. R2 = 576Ω ±5%, 1/4 W.

FIGURE 10. Output voltage (high-level and low-level, V_{DD} = 15 V) test circuit for tables IX and X.

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Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1A Input current, high level	4	3,2,5,6,25,31,9,8,34,12,11,35, 36,37,40,39,38	0	200	nA	11
K2	2A	3	4,2,5,6,25,31,9,8,34,12,11,35, 36,37,40,39,38				
K3	3A	2	4,3,5,6,25,31,9,8,34,12,11,35, 36,37,40,39,38				
K4	4A	5	4,3,2,6,25,31,9,8,34,12,11,35, 36,37,40,39,38				
K5	5A	6	4,3,2,5,25,31,9,8,34,12,11,35, 36,37,40,39,38				
K6	6A	25	4,3,2,5,6,31,9,8,34,12,11,35, 36,37,40,39,38				
K7	7A	31	4,3,2,5,6,25,9,8,34,12,11,35, 36,37,40,39,38				
K8	8A	9	4,3,2,5,6,25,31,8,34,12,11,35, 36,37,40,39,38				
K9	9A	8	4,3,2,5,6,25,31,9,34,12,11,35, 36,37,40,39,38				
K10	10A	34	4,3,2,5,6,25,31,9,8,12,11,35, 36,37,40,39,38				
K11	11A	12	4,3,2,5,6,25,31,9,8,34,11,35, 36,37,40,39,38				
K12	12A	11	4,3,2,5,6,25,31,9,8,34,12,35, 36,37,40,39,38				
K13	13A	35	4,3,2,5,6,25,31,9,8,34,12,11, 36,37,40,39,38				
K14	14A	36	4,3,2,5,6,25,31,9,8,34,12,11, 35,37,40,39,38				
K15	15A	37	4,3,2,5,6,25,31,9,8,34,12,11, 35,36,40,39,38				
K16	16A	40	4,3,2,5,6,25,31,9,8,34,12,11, 35,36,37,39,38				
K17	17A	39	4,3,2,5,6,25,31,9,8,34,12,11, 35,36,37,40,38				
K18	18A	38	4,3,2,5,6,25,31,9,8,34,12,11, 35,36,37,40,39				

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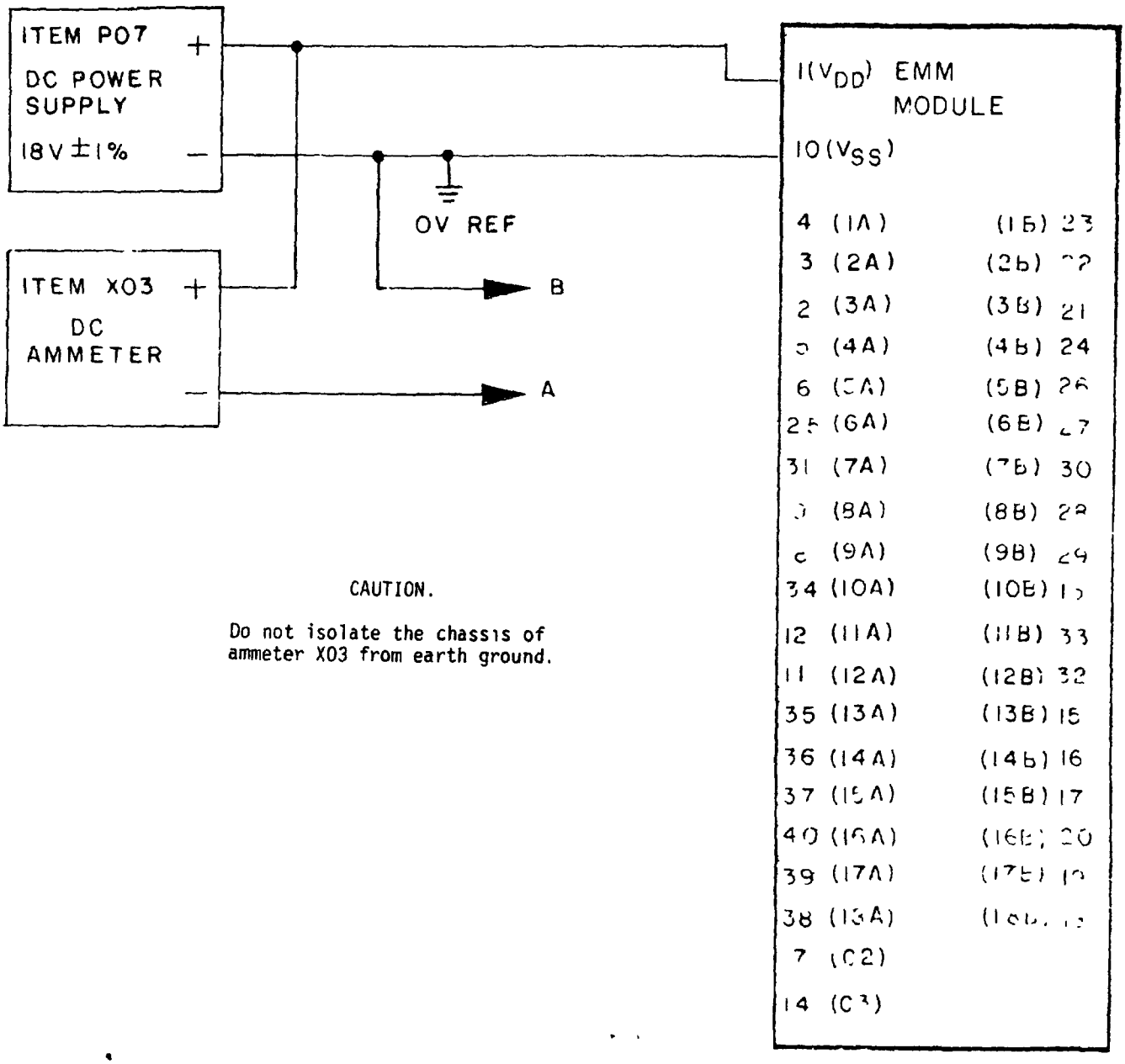


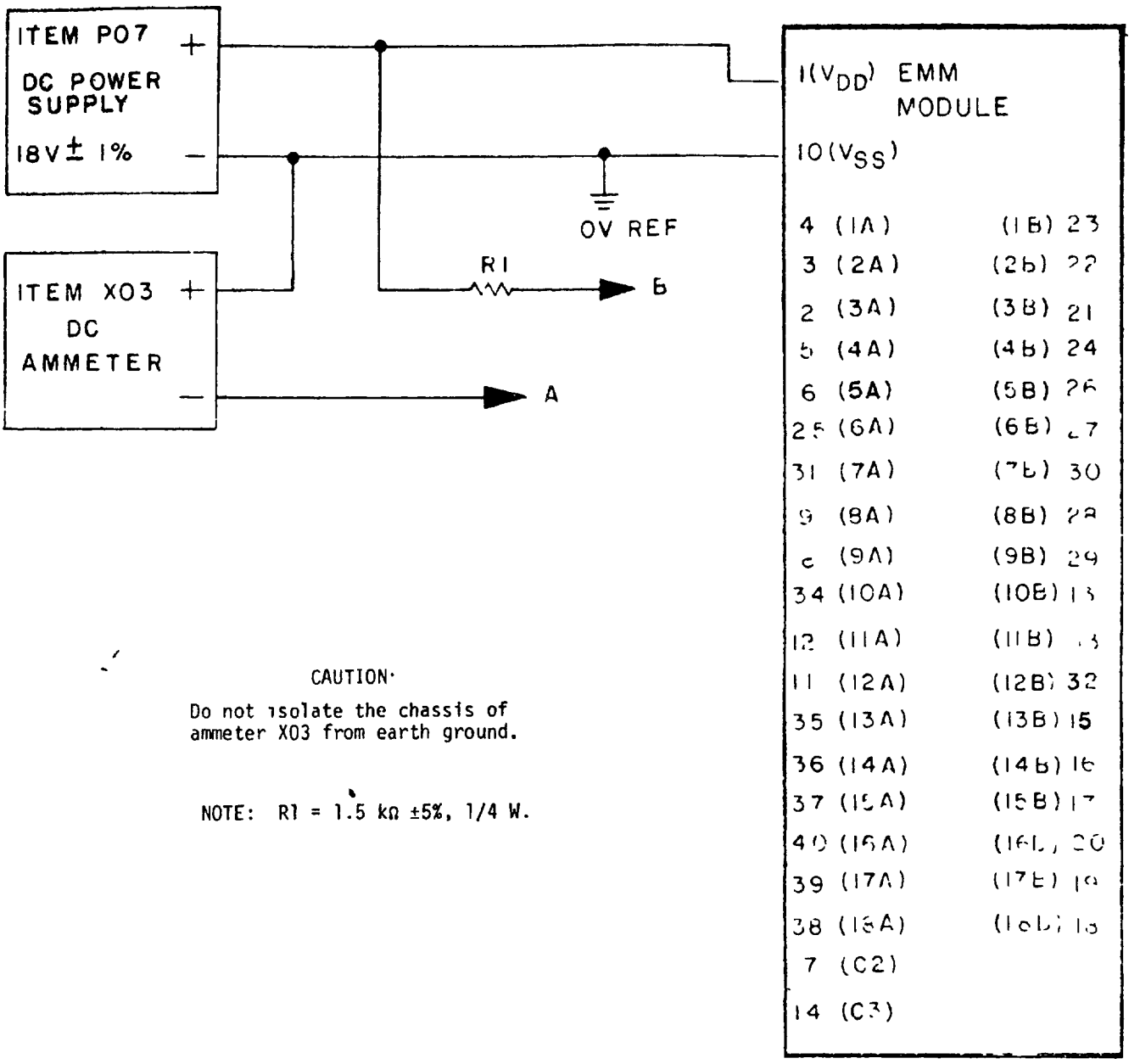
FIGURE 11. Input current (high-level) test circuit for table XI.

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TABLE XII Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	1A Input current, low level	4	3,2,5,6,25,31,9,8,34,12,11,35,36,37,40,39,38	0	-200	nA	12
L2	2A	3	4,2,5,6,25,31,9,8,34,12,11,35,36,37,40,39,38				
L3	3A	2	4,3,5,6,25,31,9,8,34,12,11,35,36,37,40,39,38				
L4	4A	5	4,3,2,6,25,31,9,8,34,12,11,35,36,37,40,39,38				
L5	5A	6	4,3,2,5,25,31,9,8,34,12,11,35,36,37,40,39,38				
L6	6A	25	4,3,2,5,6,31,9,8,34,12,11,35,36,37,40,39,38				
L7	7A	31	4,3,2,5,6,25,9,8,34,12,11,35,36,37,40,39,38				
L8	8A	9	4,3,2,5,6,25,31,8,34,12,11,35,36,37,40,39,38				
L9	9A	8	4,3,2,5,6,25,31,9,34,12,11,35,36,37,40,39,38				
L10	10A	34	4,3,2,5,6,25,31,9,8,12,11,35,36,37,40,39,38				
L11	11A	12	4,3,2,5,6,25,31,9,8,34,11,35,36,37,40,39,38				
L12	12A	11	4,3,2,5,6,25,31,9,8,34,12,35,36,37,40,39,38				
L13	13A	35	4,3,2,5,6,25,31,9,8,34,12,11,36,37,40,39,38				
L14	14A	36	4,3,2,5,6,25,31,9,8,34,12,11,35,37,40,39,38				
L15	15A	37	4,3,2,5,6,25,31,9,8,34,12,11,35,36,40,39,38				
L16	16A	40	4,3,2,5,6,25,31,9,8,34,12,11,35,36,37,39,38				
L17	17A	39	4,3,2,5,6,25,31,9,8,34,12,11,35,36,37,40,38				
L18	18A	38	4,3,2,5,6,25,31,9,8,34,12,11,35,36,37,40,39				

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CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

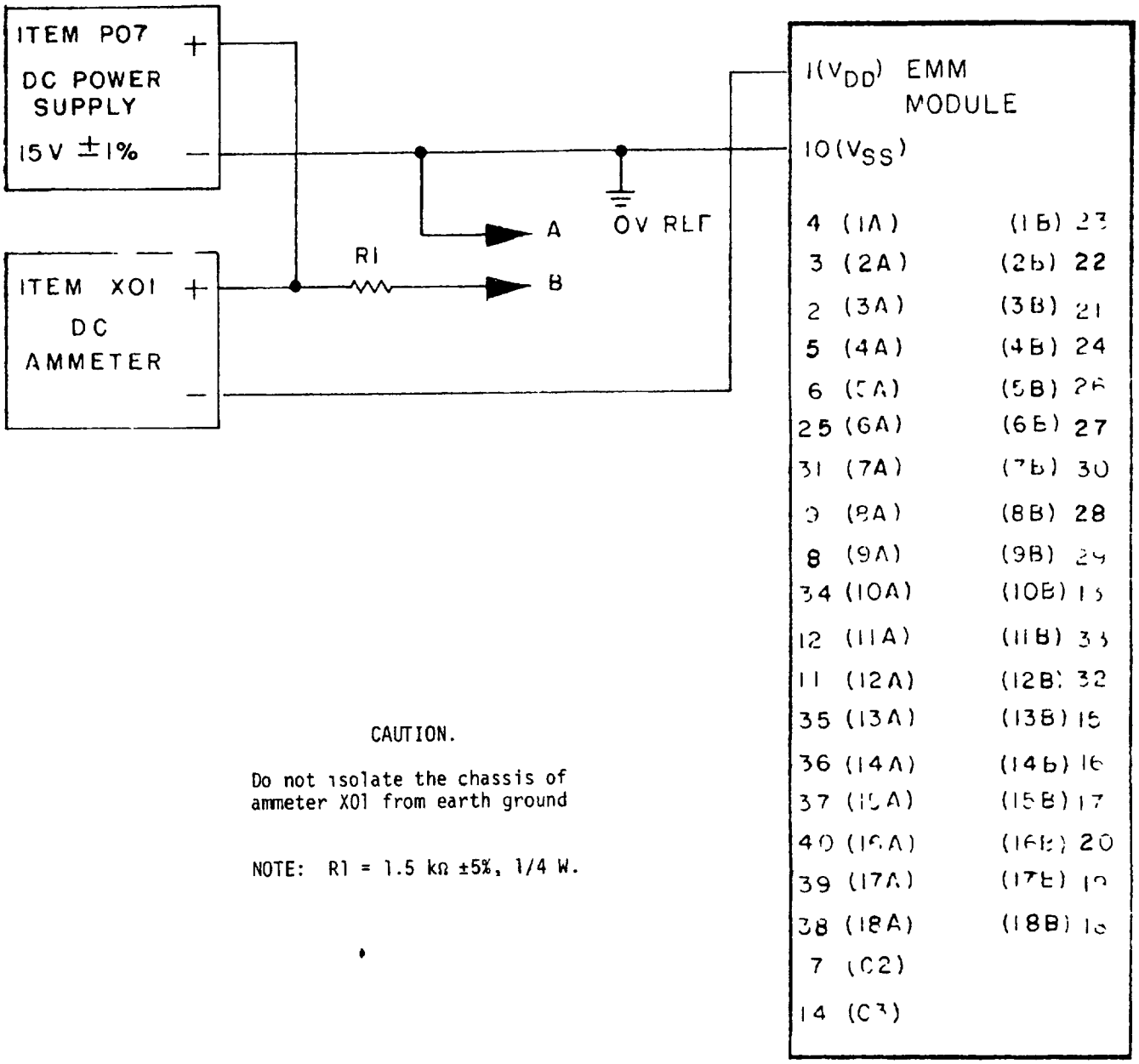
FIGURE 12. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	Open	4,3,2,5,6,25,31,9,8,34,12,11,35,36,37,40,39,38	0	6	µA	13
M2	Power supply current, high level	4,3,2,5,6,25,31,9,8,34,12,11,35,36,37,40,39,38	Open	0	6	µA	13

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CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIII.

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TABLE XIV Transition time (t_{TLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
N1	1B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	20	200	ns	14,15
N2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
N3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
N4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
N5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
N6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
N7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
N8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
N9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
N10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
N11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
N12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XIV. Transition time (t_{TLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N13	13B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	20	200	ns	14,15
N14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
N15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
N16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
N17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
N18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				
N19	1B Transition time (t_{TLH}), C_l = 150 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	68	680		
N20	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
N21	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
N22	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
N23	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
N24	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				

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TABLE XIV. Transition time (t_{TLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N25	7B Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30	68	680	ns	14,15
N26	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
N27	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
N28	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
N29	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
N30	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				
N31	13B	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15				
N32	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
N33	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
N34	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
N35	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
N36	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XV. Transition time (t_{THL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	1B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	20	200	ns	14,15
P2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
P3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
P4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
P5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
P6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
P7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
P8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
P9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
P10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
P11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
P12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XV. Transition time (t_{THL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P13	13B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	20	200	ns	14,15
P14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
P15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
P16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
P17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
P18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				
P19	1B Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	68	680		
P20	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
P21	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
P22	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
P23	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
P24	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				

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TABLE XV. Transition time (t_{THL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P25	7B Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30	68	680	ns	14,15
P26	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
P27	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
P28	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
P29	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
P30	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				
P31	13B	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15				
P32	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
P33	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
P34	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
P35	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
P36	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XVI. Transition time (t_{TLH}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	10	100	ns	14,15
Q2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
Q3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
Q4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
Q5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
Q6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
Q7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
Q8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
Q9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
Q10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
Q11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
Q12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XVI. Transition time (t_{TLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q13	13B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	10	100	ns	14,15
Q14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
Q15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
Q16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
Q17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
Q18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XVII. Transition time (t_{THL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	10	100	ns	14,15
R2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
R3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
R4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
R5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
R6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
R7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
R8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
R9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
R10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
R11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
R12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XVII. Transition time (t_{THL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R13	13B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	10	100	ns	14,15
R14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
R15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
R16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
R17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36 ,40,38	19				
R18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XVIII. Transition time (t_{TLH}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	8	80	ns	14,15
S2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
S3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
S4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
S5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
S6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
S7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
S8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
S9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
S10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
S11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
S12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XVIII. Transition time (t_{TLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S13	13B Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	8	80	ns	14,15
S14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16	↓	↓	↓	↓
S15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17	↓	↓	↓	↓
S16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20	↓	↓	↓	↓
S17	17B	39	4,3,2,5,6,25,31,9,8,34 12,11,35,36,37,40,38	19	↓	↓	↓	↓
S18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18	↓	↓	↓	↓

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TABLE XIX. Transition time (t_{THL}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	8	80	ns	14,15
T2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
T3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
T4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
T5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
T6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
T7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
T8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
T9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
T10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
T11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
T12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XIX. Transition time (t_{THL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T13	13B Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	8	80	ns	14,15
T14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
T15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
T16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
T17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
T18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	1B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	12	125	ns	14,15
U2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
U3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
U4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
U5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
U6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
U7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
U8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
U9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
U10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
U11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
U12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U13	13B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	12	125	ns	14,15
U14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
U15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
U16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
U17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
U18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				
U19	1B Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	32	325		
U20	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
U21	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
U22	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
U23	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				

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TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U24	6B Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27	32	325	ns	14,15
U25	/B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
U26	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
U27	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
U28	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
U29	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
U30	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				
U31	13B	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15				
U32	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
U33	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
U34	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
U35	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
U36	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	1B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	12	125	ns	14,15
V2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
V3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
V4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
V5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
V6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
V7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
V8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
V9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
V10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
V11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
V12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V13	13B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	12	125	ns	14,15
V14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
V15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
V16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
V17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
V18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				
V19	1B Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	32	325		
V20	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
V21	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
V22	4B	5	4,3,2,6,25,31,9,8,34, 12,11,35,36,37,40,39,38	24				
V23	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
V24	6B	25	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39,38	27				

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TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V25	7B Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30	32	325	ns	14,15
V26	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
V27	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
V28	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
V29	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
V30	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				
V31	13B	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15				
V32	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
V33	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
V34	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
V35	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
V36	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XXII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	8	80	ns	14,15
W2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
W3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
W4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
W5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
W6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
W7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
W8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
W9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
W10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
W11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
W12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XXII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W13	13B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	8	80	ns	14,15
W14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
W15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
W16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
W17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
W18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XXIII. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	8	80	ns	14,15
X2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
X3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
X4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
X5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
X6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
X7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
X8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
X9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
X10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
X11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
X12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XXIII. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X13	13B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	8	80	ns	14,15
X14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
X15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
X16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
X17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
X18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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TABLE XXIV. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y1	1B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	6	60	ns	14,15
Y2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
Y3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
Y4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
Y5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
Y6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
Y7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
Y8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
Y9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
Y10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
Y11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
Y12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XXIV. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y13	13B Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	6	60	ns	14,15
Y14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
Y15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
Y16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
Y17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
Y18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

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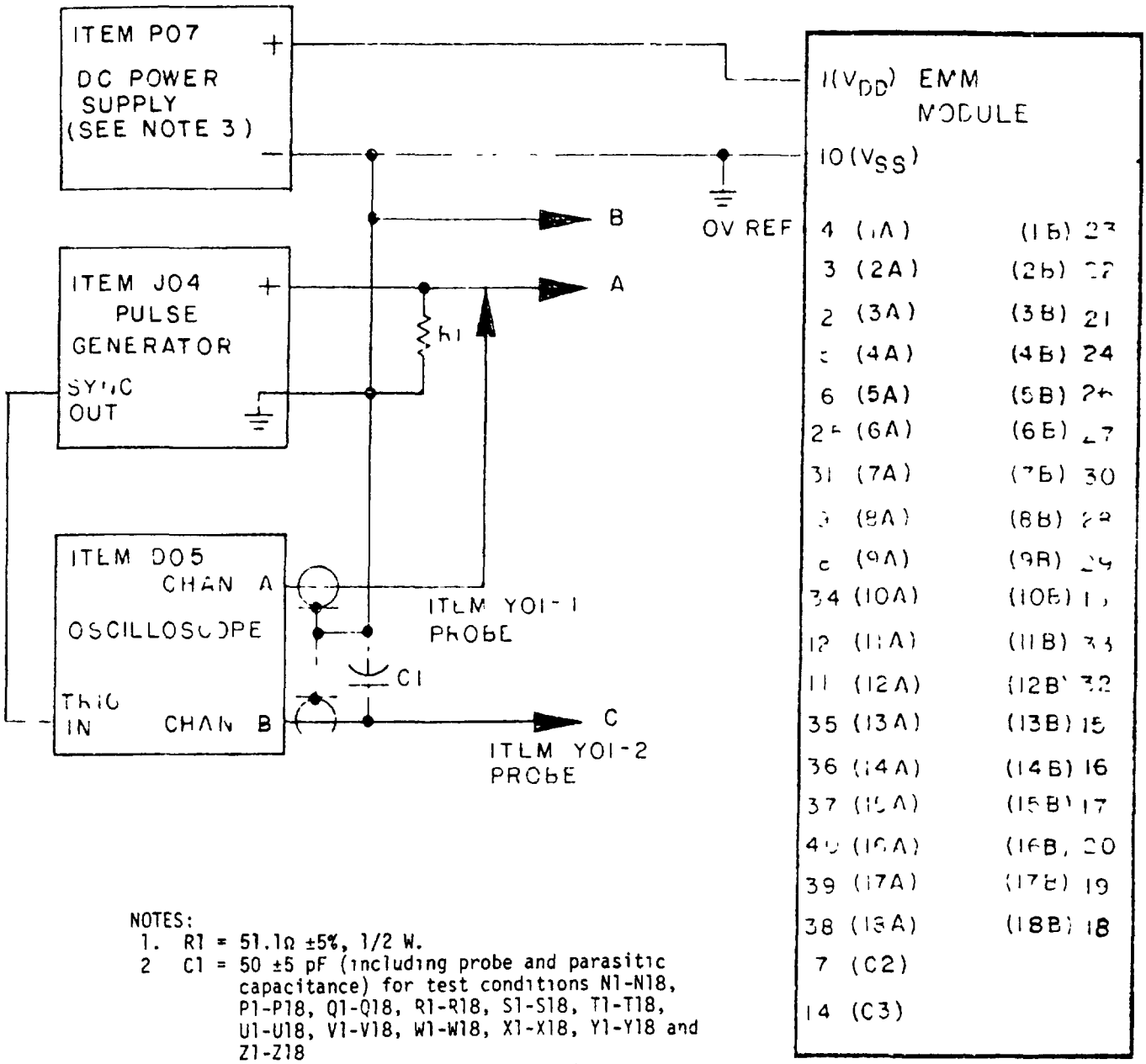
Test condition	Parameter	Terminal conditions A to pin	B to pin	C to pin	Limits 25°±5°C ambient Min	Limits 25°±5°C ambient Max	Unit	Test figure
Z1	1B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	4	3,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	23	6	60	ns	14,15
Z2	2B	3	4,2,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	22				
Z3	3B	2	4,3,5,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	21				
Z4	4B	5	4,3,2,6,25,31,9,8,34,12, 11,35,36,37,40,39,38	24				
Z5	5B	6	4,3,2,5,25,31,9,8,34,12, 11,35,36,37,40,39,38	26				
Z6	6B	25	4,3,2,5,6,31,9,8,34,12, 11,35,36,37,40,39,38	27				
Z7	7B	31	4,3,2,5,6,25,9,8,34,12, 11,35,36,37,40,39,38	30				
Z8	8B	9	4,3,2,5,6,25,31,8,34,12, 11,35,36,37,40,39,38	28				
Z9	9B	8	4,3,2,5,6,25,31,9,34,12, 11,35,36,37,40,39,38	29				
Z10	10B	34	4,3,2,5,6,25,31,9,8,12, 11,35,36,37,40,39,38	13				
Z11	11B	12	4,3,2,5,6,25,31,9,8,34, 11,35,36,37,40,39,38	33				
Z12	12B	11	4,3,2,5,6,25,31,9,8,34, 12,35,36,37,40,39,38	32				

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TABLE XXV. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z13	13B Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	35	4,3,2,5,6,25,31,9,8,34, 12,11,36,37,40,39,38	15	6	60	ns	14,15
Z14	14B	36	4,3,2,5,6,25,31,9,8,34, 12,11,35,37,40,39,38	16				
Z15	15B	37	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,40,39,38	17				
Z16	16B	40	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,39,38	20				
Z17	17B	39	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,38	19				
Z18	18B	38	4,3,2,5,6,25,31,9,8,34, 12,11,35,36,37,40,39	18				

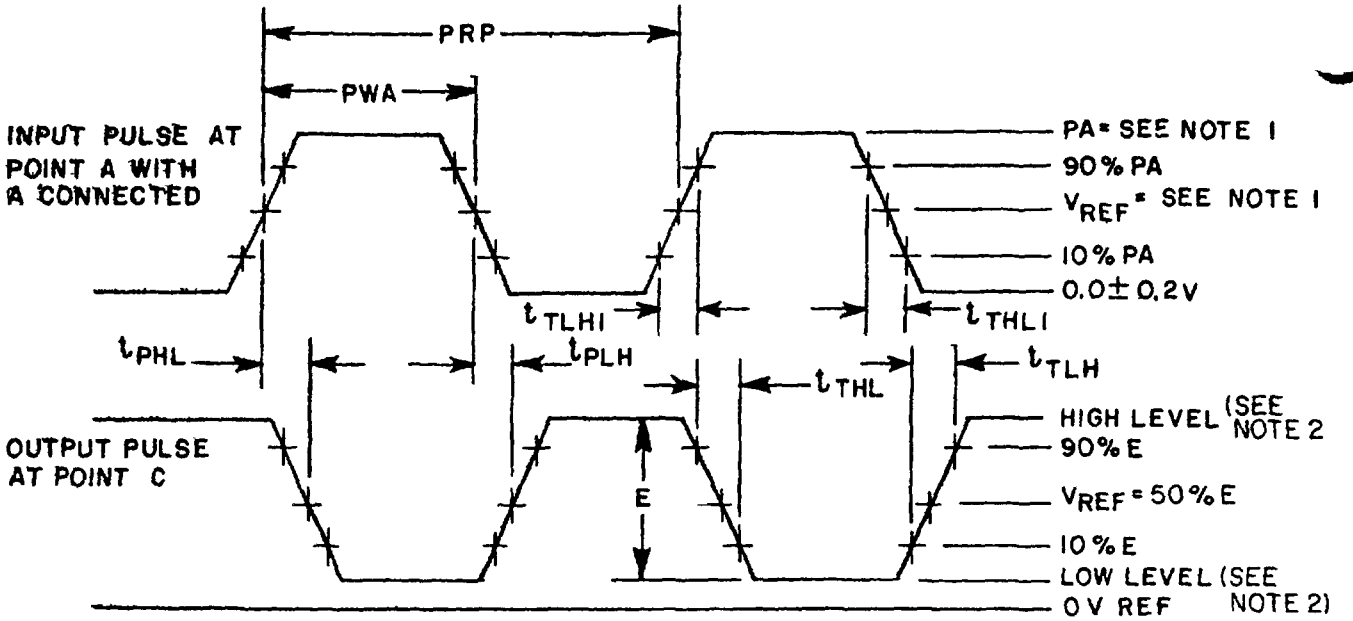
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N18, P1-P18, Q1-Q18, R1-R18, S1-S18, T1-T18, U1-U18, V1-V18, W1-W18, X1-X18, Y1-Y18 and Z1-Z18
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions N19-N36, P19-P36, U19-U36, and V19-V36
3. Set item P07 as follows
   5 V ±1% for test conditions N1-N36, P1-P36, U1-U36 and V1-V36
   10 V ±1% for test conditions Q1-Q18, R1-R18, W1-W18 and X1-X18
   15 V ±1% for test conditions S1-S18, T1-T18, Y1-Y18 and Z1-Z18

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIV through XXV.

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NOTES:

1. Input pulse parameters
   - PRP = 5.0 ±0.5 µs.
   - $t_{TLHI} = t_{THLI}$ = 10 ±2 ns.
   - PWA = 2.5 ±0.2 µs for test conditions N1-N36, P1-P36, U1-U36 and V1-V36.
     = 1.0 ±0.1 µs for test conditions Q1-Q18, R1-R18, S1-S18, T1-T18, W1-W18, X1-X18, Y1-Y18 and Z1-Z18.
   - PA = 5.0 ±0.1 V for test conditions N1-N36, P1-P36 and U1-U36 and V1-V36.
     = 10.0 ±0.1 V for test conditions Q1-Q18, R1-R18, W1-W18 and X1-X18
     = 15.0 ±0.1 V for test conditions S1-S18, T1-T18, Y1-Y18 and Z1-Z18.
   - V_{REF} = 2.5 ±0.1 V for test conditions N1-N36, P1-P36, U1-U36 and V1-V36
     = 5.0 ±0.1 V for test conditions Q1-Q18, R1-R18, W1-W18 and X1-X18
     = 7.5 ±0.1 V for test conditions S1-S18, T1-T18, Y1-Y18 and Z1-Z18
2. Before making the measurements, verify that the output pulse high level and low level is as follows
   - High level ≥ 4.6 V for test conditions N1-N36, P1-P36, U1-U36 and V1-V36
     ≥ 9.5 V for test conditions Q1-Q18, R1-R18, W1-W18 and X1-X18.
     ≥ 13.5 V for test conditions S1-S18, T1-T18, Y1-Y18 and Z1-Z18
   - Low level ≤ 0.4 V for test conditions N1-N36, P1-P36, U1-U36 and V1-V36.
     ≤ 0.5 V for test conditions Q1-Q18, R1-R18, W1-W18 and X1-X18.
     ≤ 1.5 V for test conditions S1-S18, T1-T18, Y1-Y18 and Z1-Z18
3. The lead length of electrical conductors between resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins, and between load circuit and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground (V_{SS}).

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIV through XXV.

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TABLE XXVI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	0.007	0.013	µF	16

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TABLE XXVII. C2 and C3 capacitance test.

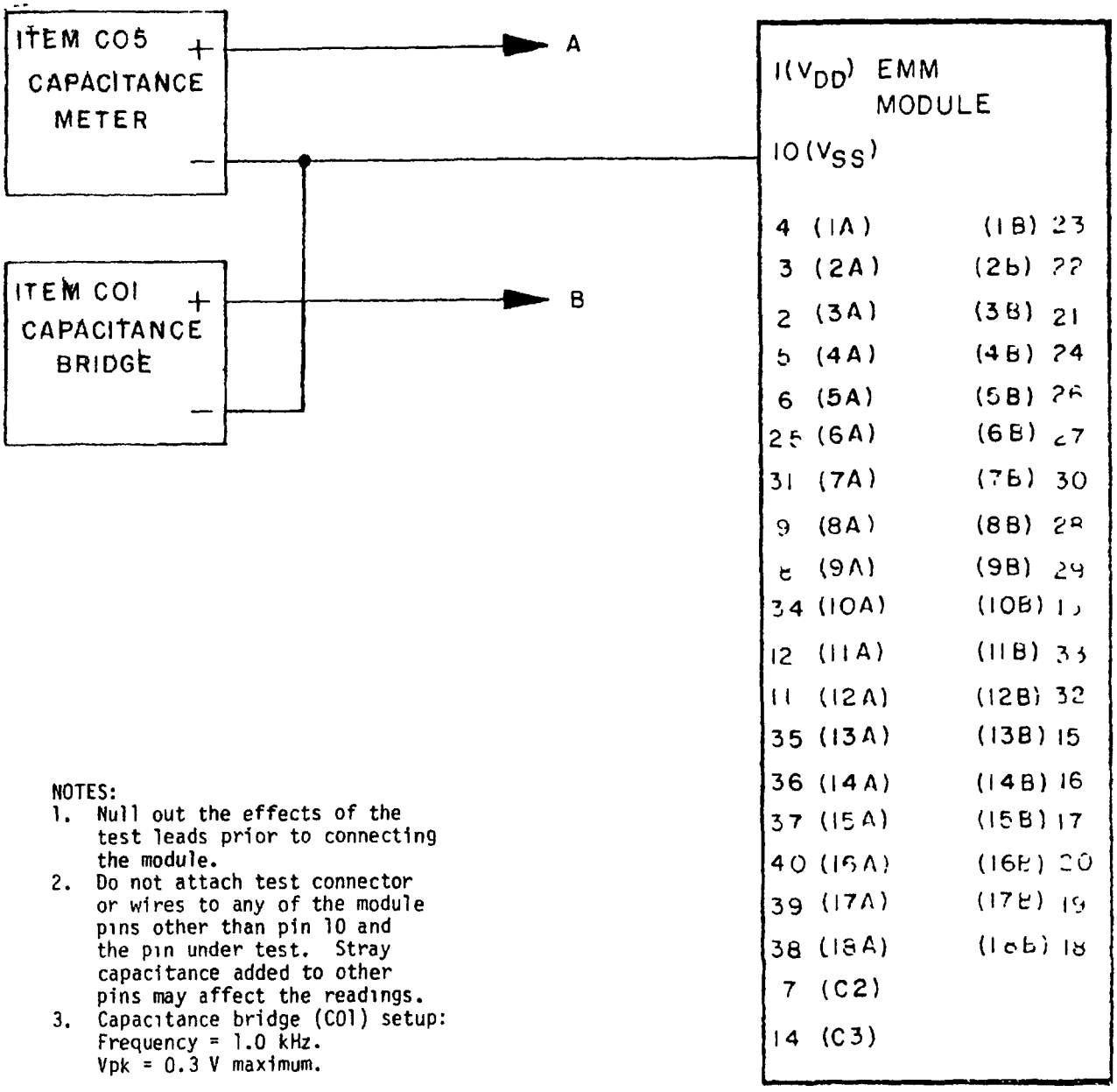
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	C2 capacitance	7	Open	176	224	pF	16
BB2	C3 capacitance	14	Open	176	224	pF	16

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TABLE XXVIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1A Input capacitance	4	Open	2	15	pF	16
CC2	2A	3					
CC3	3A	2					
CC4	4A	5					
CC5	5A	6					
CC6	6A	25					
CC7	7A	31					
CC8	8A	9					
CC9	9A	8					
CC10	10A	34					
CC11	11A	12					
CC12	12A	11					
CC13	13A	35					
CC14	14A	36					
CC15	15A	37					
CC16	16A	40					
CC17	17A	39					
CC18	18A	38					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 1.0 kHz.
   Vpk = 0.3 V maximum.

FIGURE 16. Power supply filter, C2 and C3, and input capacitance test circuit for tables XXVI, XXVII and XXVIII.

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